

Q1:)

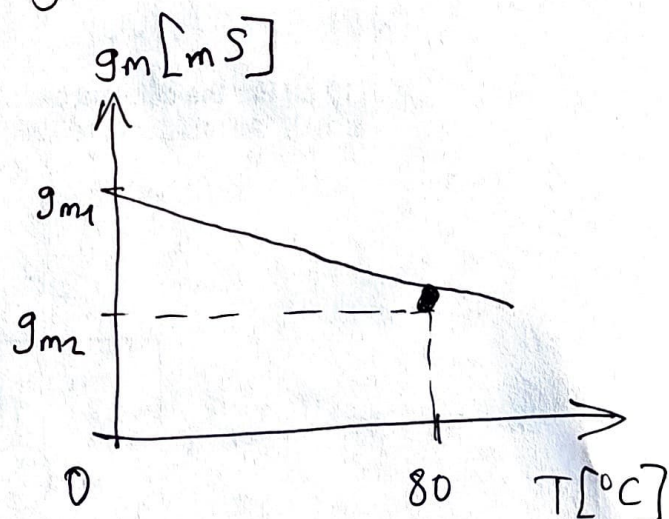
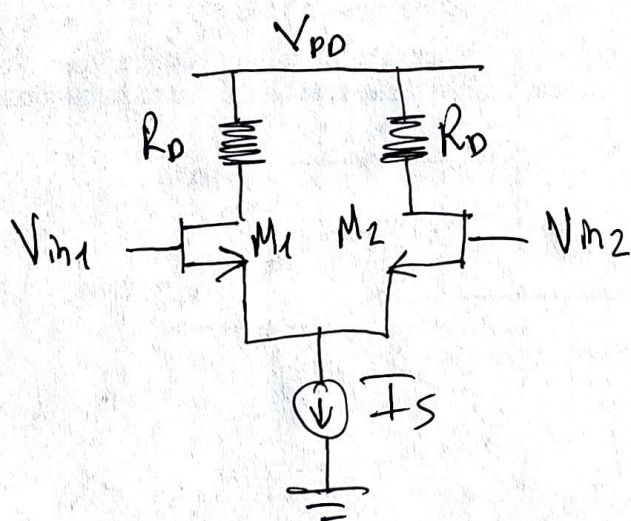
①

A differential amplifier is built using two identical nmos transistors and two identical drain resistors $R_D = 10k\Omega$. The transistors operate in saturation region with a constant bias current. The gain of the amplifier is:

$$A_v = -g_m R_D \text{ or } |A_v| = g_m R_D \rightarrow \text{small-signal gain}$$

Assume that the mobility of the nmos transistors depends on temperature as

$$\mu_n \propto T^{-1.5} \text{ and } g_m \propto \mu_n \Rightarrow g_m \propto T^{-0.75}$$



Let the amplifier operate at two different temperatures: $T_1 = 273\text{ K}$ and $T_2 = 353\text{ K}$. Assume, the drain current remains constant with temperature:

a) Calculate the ratio g_{m2}/g_{m1} , where g_{m1} and g_{m2} are the transconductances at T_1 and T_2 , respectively. What percentage does g_m decrease by over this temperature range?

$$\frac{g_{m2}}{g_{m1}} = \left(\frac{T_2}{T_1} \right)^{-0.75} = \left(\frac{353}{273} \right)^{-0.75}$$

(2)

$$\frac{353}{273} \approx 1.293 \Rightarrow (1.293)^{-0.75} \approx \boxed{0.8245}$$

The transconductance decreases by 17.55%.

b) The transistors are biased with a tail current source I_S , which is designed to have a PTAT characteristic. If the current source is designed to be PTAT, determine what temperature dependence the tail current must have in order for the differential-mode gain A_v to remain constant across temperature. Express your answer in the form $I_D(T) \propto T^X$. Determine the value of the exponent X .

$$g_m(T) \propto \sqrt{I_{n1}(T) \cdot I_{D1}(T)} = \sqrt{T^{-1.5} \cdot I_{D1}(T)} \text{ and}$$

$$\sqrt{T^{-1.5} \cdot I_{D1}(T)} = \text{const} \Rightarrow \boxed{I_{D1}(T) \propto T^{1.5}}$$

c) At $T = 300\text{K}$, the tail current is $I_D(300\text{K}) = 100\mu\text{A}$ and the gain is measured as $A_v = -20$. What is the value of g_m at 300K ? What should the tail current $I_S(360\text{K})$ be in order to maintain the same gain?

$$A_v = -g_m \cdot R_D = -20 \text{ and } R_D = 10\text{k}\Omega$$

$$g_m = \frac{-A_v}{R_D} = \underline{\underline{2\text{mS}}}$$

$$g_m(T) \propto \sqrt{I_{M(T)} I_{D(T)}} = \sqrt{T^{-1.5} \cdot I_{D(T)}} \quad (3)$$

to maintain the same g_m/R_D or g_m , we require

$$\sqrt{T^{-1.5} \cdot I_{D(T)}} = \text{const} \rightarrow I_{D(T)} \propto T^{1.5}$$

$$\frac{I_D(360)}{I_D(300)} = \left(\frac{360}{300}\right)^{1.5} \approx 1.314$$

$$I_D(360K) = 1.314 \cdot 100 \mu A = \boxed{131.4 \mu A}$$

Q2:) Consider the CMOS compatible PTAT current generator shown in the figure. Assume the following:

- * $M_1 - M_2$ in saturation
- * $M_3 - M_4$ " "

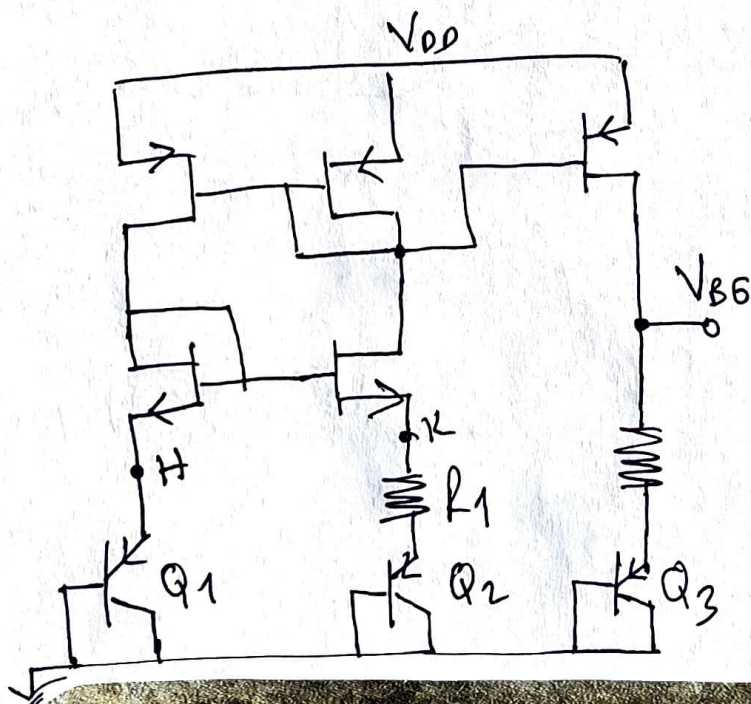
* Q_1 and Q_2 are BJTs with emitter area ratio

$$\frac{A_2}{A_1} = n = 8$$

$$R_1 = 12k\Omega, T = 300K$$

$$V_T = \frac{kT}{q} \approx 26mV$$

- * CLM neglected!
- * Current mirror is ideal



a) Derive an expression for the PTAT current I (4) generated by the circuit.

$$V_H - V_L = |V_{BE1}| - |V_{BE2}| \rightarrow R_1 I = |V_{BE1}| - |V_{BE2}|$$

$$|V_{BE1}| - |V_{BE2}| = V_T \ln\left(\frac{A_2}{A_1}\right) = V_T \ln(n)$$

The PTAT current is:

$$I = \frac{V_T}{R_1} \ln(n)$$

$\rightarrow$ This current is proportional to absolute temperature (PTAT) and independent of V_{DD} !

b) Compute the numerical value of the PTAT current I

at $T = 300\text{K}$.

$$V_T = 26\text{mV}$$

$$R_1 = 12\text{k}\Omega$$

$$n = 8, \text{ so } \ln(8) \approx 2.079$$

$$I = \frac{26 \cdot 10^{-3}}{12 \cdot 10^3} \cdot 2.079$$

$$\boxed{I \approx 4.5\text{mA}}$$

c) Compute the voltage drop across R_1

$$V_{R1} = I \cdot R_1 = 4.5\text{mA} \cdot 12\text{k}\Omega \approx \boxed{54\text{mV}}$$

d) If the temperature rises to $T = 330\text{K}$, calculate the new PTAT current and the percentage increase in current

$$V_T(330) = 26\text{mV} \cdot \frac{330}{300} \approx 28.6\text{mV}$$

$$I_{330} = \frac{28.6 \cdot 10^{-3}}{12 \cdot 10^3} \cdot 2.079 \approx \boxed{4.95\text{mA}}$$

$$\boxed{\Delta I \approx 10\%}$$